



Varendra University
Department of Computer Science and Engineering

Undergraduate Project Proposal

Title of the Project

Blockchain-Based Secure Academic Certificate Verification System

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1. Introduction

The rapid growth of digital education and online academic services has increased the risk of forged and manipulated academic certificates. Traditional certificate verification methods rely on manual checking or centralized databases, which are time-consuming, vulnerable to tampering, and dependent on institutional trust. Fake certificates can cause serious consequences in recruitment, higher education admissions, and professional credentialing.

Blockchain is a technology that allows information to be stored securely. The main advantage of blockchain is that once information is stored, it cannot be changed or removed without leaving a trail. This means it is virtually impossible to hack information stored in a blockchain. The combination of a smart contract and a cryptographic hash is what gives blockchain the potential to solve the problem of counterfeit academic certificates.

This project proposes a Blockchain-Based Secure Academic Certificate Verification System built using Ethereum smart contracts. Rather than storing entire certificate files on the blockchain, the system generates a unique SHA-256 hash for each certificate and stores that hash immutably. When a certificate needs to be verified, the system hashes the uploaded file and compares it with the stored blockchain record to confirm its authenticity. The goal of this project is to develop a secure, transparent, and cost-effective prototype for reliable academic credential verification.

2. Literature Review / Background Study

Several studies have explored the use of blockchain in academic credential management.

Sharples and Domingue [1] proposed a blockchain-based system for storing educational achievements to ensure transparency and integrity. Their work demonstrated how blockchain can eliminate centralized verification dependencies but highlighted scalability concerns.

Grech and Camilleri [2] analyzed blockchain applications in education and identified credential verification as a major use case. However, they noted high implementation complexity and infrastructure cost.

Turkanović et al. [3] introduced a blockchain-based diploma verification platform that stores academic records securely. Their system provided immutability but required complex deployment and public blockchain usage.

The Ethereum whitepaper by Buterin [4] introduced smart contracts, enabling programmable blockchain logic suitable for certificate validation systems.

Nakamoto [5] first introduced the concept of blockchain through Bitcoin, describing a decentralized and tamper-resistant ledger system secured by cryptographic hashing. This

foundational work established the core principles of immutability, transparency, and trustless verification that modern blockchain-based credential systems rely upon.

While existing systems demonstrate feasibility, many rely on public blockchain deployment with high gas fees or complex architectures. This project proposes a simplified and cost-effective prototype using a local Ethereum test environment to achieve secure verification with minimal complexity.

3. Problem Statement

The problem of academic certificate forgery is increasingly becoming an issue of concern for recruitment and institutions. Currently, verification mechanisms used are often manual and time-consuming, depending on data from centralized sources that can be tampered with by authorized personnel or attackers. Therefore, there is a need to develop a tamper-evident and transparent certificate verification mechanism that ensures certificate authenticity without depending on centralized trust.

This project aims to address this problem by implementing a blockchain-based certificate verification system that ensures immutability and integrity using cryptographic hashing and smart contracts.

4. Objectives

- To design and implement a blockchain-based certificate verification prototype within four months.
- To develop a smart contract that securely stores certificate hashes.
- To create a web-based interface for certificate upload and verification.
- To evaluate system functionality and integrity through testing scenarios.

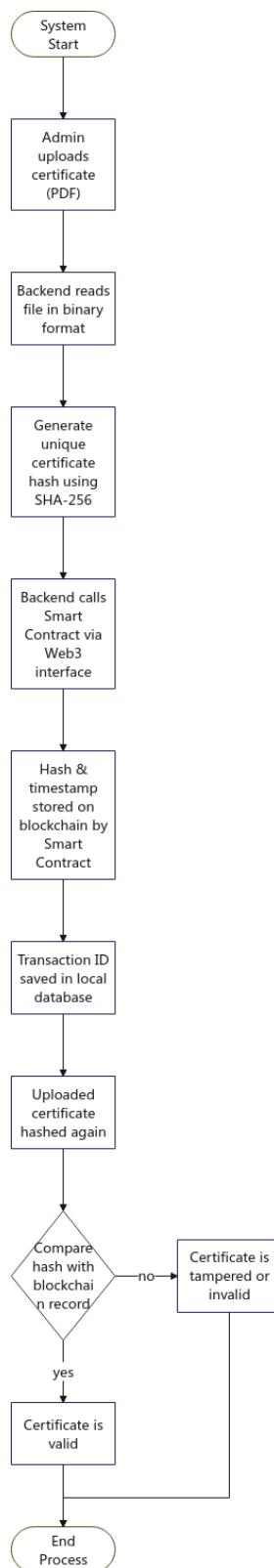
5. Proposed Methodology / System Design

The proposed system follows a layered architecture consisting of frontend, backend, database, smart contract, and blockchain network.

System Flow:

1. Admin uploads certificate (PDF).
2. Backend reads file in binary format.
3. SHA-256 hashing algorithm generates a unique certificate hash.
4. Backend uses Web3 interface to call the smart contract.
5. Smart contract stores hash and timestamp on the blockchain.
6. Transaction ID is saved in the local database.

7. For verification, the uploaded certificate is hashed again.
8. The system compares the generated hash with the blockchain record.
9. If matched → certificate is valid; otherwise → tampered or invalid.



The smart contract is written in Solidity and deployed on a local Ethereum test network using Ganache. Wallet interaction is managed through MetaMask.

Only certificate hashes are stored on blockchain to minimize storage cost and improve efficiency.

6. Expected Outcomes

- A fully functional web-based certificate verification system.
- Deployed smart contract on Ethereum test environment.
- Successful tamper detection demonstration.
- Technical documentation and system evaluation report.

7. Tools and Technologies to Be Used

- Python
- Flask Framework
- Solidity
- Ethereum
- Ganache
- MetaMask
- Web3.py
- SHA-256 hashing
- MySQL / SQLite
- HTML, CSS, JavaScript

8. References

[1] M. Sharples and J. Domingue, "The Blockchain and Kudos: A Distributed System for Educational Record, Reputation and Reward," 2016.

https://link.springer.com/content/pdf/10.1007/978-3-319-45153-4_48.pdf

[2] A. Grech and A. F. Camilleri, "Blockchain in Education," Joint Research Centre (JRC), European Commission, 2017.

https://www.um.edu.mt/library/oar/bitstream/123456789/108154/1/Blockchain_in_education%282017%29.pdf

[3] M. Turkanović et al., "EduCTX: A Blockchain-Based Higher Education Credit Platform," IEEE Access, vol. 6, 2018.

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8247166>

[4] V. Buterin, "A Next-Generation Smart Contract and Decentralized Application Platform," Ethereum Whitepaper, 2014.

https://cryptorating.eu/whitepapers/Ethereum/Ethereum_white_paper.pdf

[5] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.

<https://assets.pubpub.org/d8wct41f/31611263538139.pdf>

9. Work Division and Team Management

Member	Responsibility
Member 1	Smart contract development and blockchain integration
Member 2	Backend development and hashing implementation
Member 3	Frontend development and database management

Weekly team meetings will be conducted to monitor progress.

10. Timeline

Month	Activities
Month 1	Problem finalization, requirement analysis, literature review, project planning
Month 2	System architecture design, technology selection, database schema design
Month 3	Learning and setting up development environment (Ethereum, Solidity, Ganache, Web3)
Month 4	Smart contract development and initial testing
Month 5	Backend development (certificate upload, SHA-256 hashing, blockchain integration)
Month 6	Frontend development (admin panel, verification interface)
Month 7	Integration of frontend, backend, and smart contract
Month 8	System testing (functional testing, tamper testing, performance testing)
Month 9	Optimization, bug fixing, documentation writing
Month 10	Final report preparation, presentation preparation, deployment and demonstration

11. Cost-Benefit Analysis

The project uses open-source technologies and a local Ethereum test network. No real cryptocurrency or paid infrastructure is required. Therefore, the system is highly cost-effective and economically feasible. In real-world deployment, public blockchain gas fees may apply, but for academic prototype, cost is minimal.

12. Constraints and Limitations

- Prototype limited to local blockchain environment.
- Scalability not evaluated for large institutional deployment.
- Requires internet connection for blockchain interaction.

- Security depends on proper admin authentication mechanisms.

13. Societal Impact

This system reduces academic fraud and increases trust in digital credentials. It can support universities, employers, and students by providing secure and fast verification. The project promotes transparency, digital trust, and secure educational ecosystems.

14. Complex Engineering Problem Mapping

Table 1: Mapping with complex problem solving.

WP1 Dept of Knowledge	WP2 Range of Conflicting Requirements	WP3 Depth of Analysis	WP4 Familiarity of Issues	WP5 Extent of Applicable Codes	WP6 Extent of Stakeholder Involvement	WP7 Interdependence
✓	✓	✓	✓	✓	✓	✓

15. Complex Engineering Activity Mapping

Table 2: Mapping with complex engineering activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓	✓	✓

16. Green Computing Considerations

The system stores only cryptographic hashes instead of full documents, reducing blockchain storage consumption. The use of a local test network avoids unnecessary energy consumption associated with public blockchain mining.

17. Lifelong Learning and Personal Development

This project enhances knowledge in blockchain technology, smart contract development, cryptographic security, and full-stack web integration. It strengthens teamwork, problem-solving ability, and research-based thinking, contributing to long-term professional development in emerging technologies.